Electricity Costs in Oregon Households

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ST 595 Technical Report

Introduction

The analysis in this report will aim to answer key questions conducted with a subset from the American Community Survey (ACS) PUMS dataset in Oregon from 2015. The data was filtered to include a subset of variables and only households that have at least one person pay for their electricity, and are not group accommodation. It contains detailed household-level information on housing characteristics, energy use, and demographic variables for a representative sample of U.S. households.

After the data was filtered and cleaned, the final subset contained 37 predictor variables, including BLD which tells us if the unit is an apartment, house, or other, ELEP which tells us the annual electricity cost (USD), NP which tells us the number of occupants in the home, and BDSP which holds the number of bedrooms in the household.

One question of interest we are aiming to answer is whether people living in apartments pay less for electricity than those living in houses. During the explanatory analysis we will adjust the number of bedrooms and the number of occupants in the household. The second question of interest will examine creating a model that could be used to predict electricity costs for a household in Oregon.

The report will discuss and outline the methods, results and discussions of both the explanatory and predictive analysis.

Methods

Do People Living in Apartments Pay Less for Electricity Than Those Living in Houses?

To explore this question, I conducted an initial exploratory data analysis to identify relevant variables. The BLD variable, which tells us if the unit is an apartment, house, or other was used to create a subset of the data that contained only apartments and houses. From there, I fit a linear regression model with all pairwise interactions between the response variable ELEP (electricity) and primary explanatory variables: HA (household type: apartment or house), NP (number of occupants), and BDSP (number of bedrooms).

To end up with a model that avoids unnecessary overfitting and explains the response variable well I then used the stepwise model selection technique called backward elimination. I refined the model by fitting a full model with all potential variables, removing variables based on p-values (least statistically significant), and re-fitting the model without that variable. I repeated the process until I was left with a reduced model that determined which variables were the most meaningful in explaining the variation in electricity costs (p-value threshold is often 0.05).

I then analyzed the residual diagnostics with residual plots to assess the assumption of normality and model validation.

The models I worked with met the necessary linear regression assumptions (linearity, independence, homoscedasticity, and normality of residuals). To determine whether the inclusion of interaction terms improved model performance, I proceeded to compare the full model with variable interactions to the reduced model without the variable interactions using the Extra Sum of Squares F-test.

Predicting Electricity Costs for a Household in Oregon

To explore the prediction problem, I created a model that could be used to predict electricity costs for a household in Oregon. I took a systematic approach by cleaning the data, identifying missing values and removing variables that cause multicollinearity, such as *SERIALNO (housing unit number)*, *TYPE (type of unit)*, and *VALP (property*

value). The redundancy that comes from highly correlated variables can make it difficult for the model to identify which variable is influencing the response variable.

The variable SERIALNO was removed since it serves as an identifier variable and not a true feature for a prediction. The variables TYPE and VALP were excluded because they were strongly correlated to the variable BDSP (number of bedrooms) that was already accounted for housing structure in the analysis.

After cleaning, I split the dataset into a training set (80%) and a validation set (20%). The training set was used to select the best subset of variables using the *regsubsets* function. Different models of varying sizes were considered based on the Bayesian Information Criterion (BIC). Cross validation was also used to measure the model's performance with k-10 folds. This helped determine the optimal number of predictor variables.

Once the predictor variable selection was completed, the Root Mean Squared Error (RMSE) was calculated to assess the predictive accuracy of the model on the validation set.

Results

Do People Living in Apartments Pay Less for Electricity?

```
Call:
lm(formula = ELEP ~ HA * BDSP + HA * NP + BDSP * NP, data = dataHouse)

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept)   36.0714     3.7110   9.720  < 2e-16 ***
HAhouse       16.4112     4.5401   3.615 0.000302 ***
BDSP          15.2978     2.0506   7.460 9.16e-14 ***
NP            11.5291     1.6195   7.119 1.14e-12 ***
HAhouse:BDSP  -3.0794     2.0789  -1.481 0.138572
HAhouse:NP     3.9170     1.4338   2.732 0.006303 **
BDSP:NP       -0.8999     0.4311  -2.087 0.036869 *
---
```

The initial model with all possible pairwise interactions between the explanatory variables did not indicate a notable effect between the variables *HA* (household type) and *BDSP* (number of bedrooms) (p-value = 0.139). The result can be seen in the diagram.

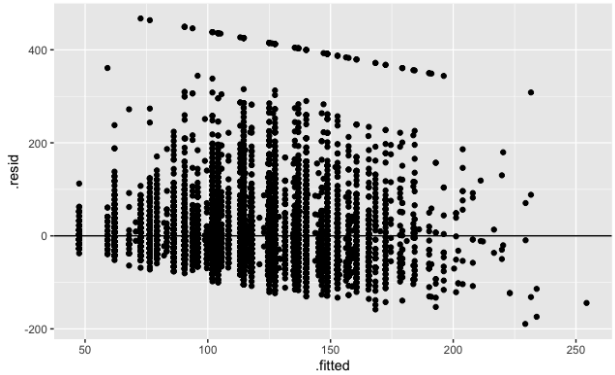
Another thing that was noted when examining the coefficient estimates of the other interaction terms, was that they had relatively small coefficient values compared to the main effects. The interaction terms between HA and NP (3.917) and BDSP and NP (-0.899), could suggest little to no contribution to the model.

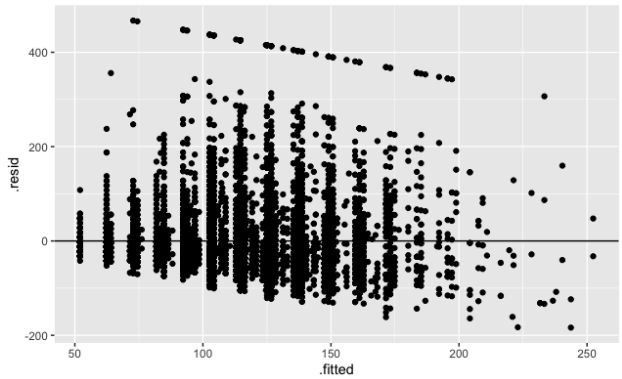
I repeated this process called backward elimination of fitting and removing terms that were not statistically significant (p-value > 0.05). Based on these results the final and reduced model did not include any of the interaction variables.

My next step after fitting both models was to check the residual diagnostics of both the full and reduced models. This will help assess and validate model assumptions of linearity, constant variance, and model fit.

Full Model

Reduced Model





In both cases, the residual vs fitted plots did not show any clear pattern. The residuals were randomly scattered around zero with a consistent spread across the fitted values. The plots suggest the assumptions of linearity and heteroscedasticity (constant variance) were not violated and the models are both a good fit for the data

To decide on the best model to use I continued to compare the full model that included all interaction terms to the reduced model which did not include the interaction terms by performing an Extra Sum of Squares F-test (Extra SS F-test). This method is useful when comparing nested models such as this case with the full and reduced models. The Extra SS F-test evaluates whether the additional terms in the full model give meaningful results and improve the model's fit to the data.

The Extra SS F-test gave a strong indication that the reduced model without the interaction terms was a better fit (p-value = 0.017). This suggests that the interaction terms did not improve the full model's fit to the data.

The reduced model showed that people living in houses pay around $19.47 more per month for electricity than people living in apartments. This difference is supported by

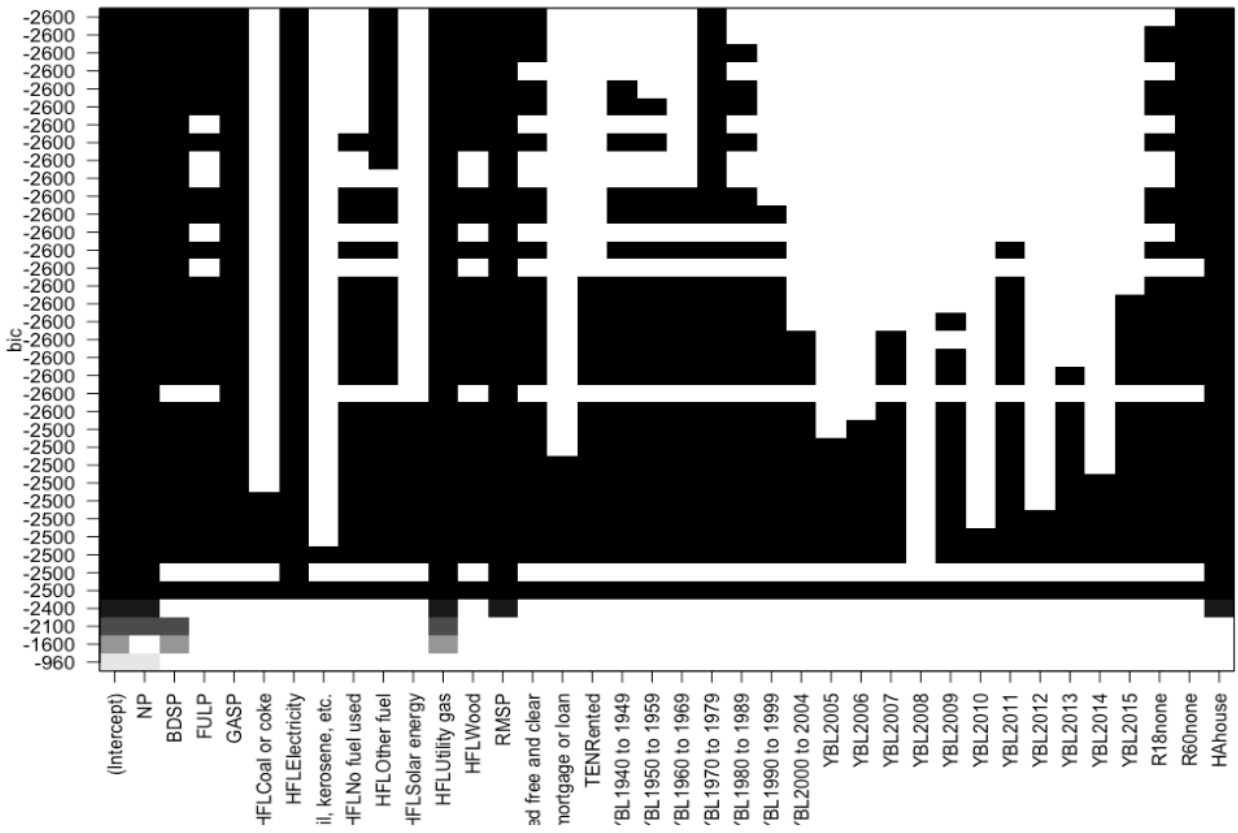
strong evidence (p-value < 0.001), including a 95% confidence interval of $15.96 and $22.99. This can be interpreted as the true difference between the cost of electricity paid by people who live in houses and people who live in apartments being between $15.96 and $22.99.

Predicting Electricity Costs for Households

I set out to create a model that could be used to predict electricity costs for a household in Oregon. The data was cleaned and split into a training set and test set.

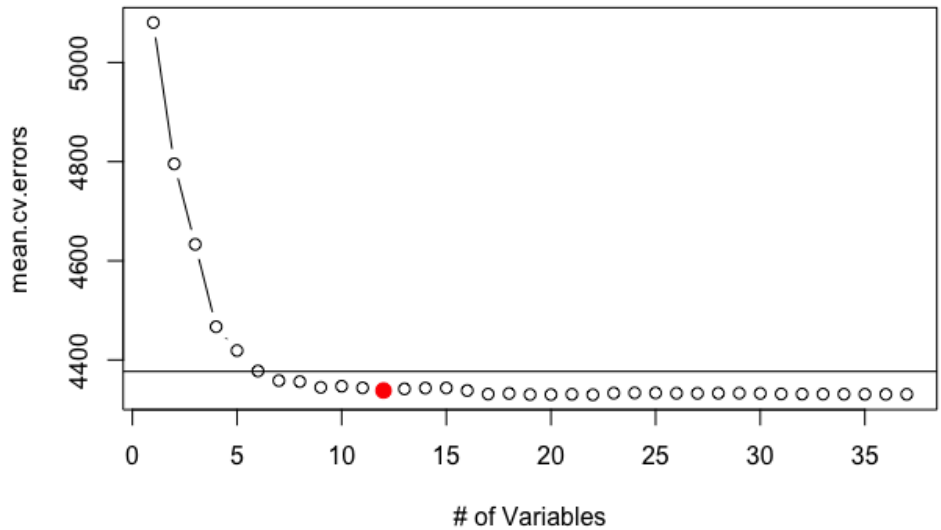
Starting with best subset selection using *regsubsets()* with BIC criteria. The built-in plot () command in regsubsets() displays the selected variables for the best model with a given number of predictor variables. The top row of the variable selection plot corresponds to a model that contains a black square for each predictor variable used. The model with 13 predictor variables, chosen for its smallest BIC, was found to provide the best fit.

Variable Selection Plot



Cross validation was also used to evaluate model performance with k-10 folds. I created a matrix in which I stored the results of the mean square errors (MSE) to perform the best subset selection within each training set. Once the MSEs were calculated they were stored in a vector, which I could then plot.

MSE Variable Selection Plot



The resulting plot indicates a model with twelve variables yields the lowest MSE value. Based on this I could then perform the best subset selection on the full data to obtain the twelve predictor variables. These predictors included variables such as NP(number of occupants), BDSP(number of bedrooms), FULP(fuel type), GASP(gas type), and heating types HFLElectricity and HFLOther.

The prediction model I fit based on the full data subset selection can be expressed as:

$$\mu\{\text{ELEP}\} = \beta_0 + \beta_1\text{NP} + \beta_2\text{BDSP} + \beta_3\text{FULP} + \beta_4\text{GASP} + \beta_5\text{RMSP} + \text{HFL} + \text{R60} + \text{HA}$$

I assessed whether including interaction terms then comparing the RMSE values of both models would result in a statistically meaningful performance increase. The model with interaction terms between NP and BDSP, and between HFL and RMSP. I estimated the Root Mean Squared Error of $66.89. Even though it is a slight decrease from the base model, that would suggest better model performance since it indicates the predictions are closer to the actual electricity costs.

Taking it a step further I compared the models by conducting an analysis of variance (ANOVA). The ANOVA results included an F-statistic of 7.404 and a p-value < 0.001. This suggests that the model that includes the interaction terms reduces error 7.404 times more than the model which does not include them. The small p-value also makes the interaction terms statistically meaningful.

The prediction model I fit based on the full data subset selection with interaction terms can be expressed as:

$$\mu\{ELEP\} = \beta 0 + \beta 1NP + \beta 2BDSP + \beta 3(NP \times BDSP) + \beta 4FULP + \beta 5GASP + \beta 6HFL + \beta 7RMSP + \beta 8(HFL \times RMSP) + \beta 9R60 + \beta 10HA$$

This linear regression model will predict electricity costs (ELEP) for households in Oregon. After fitting this model, I estimated the Root Mean Squared Error of $66.89 a month for a household in Oregon. This helps measure how well the predicted value matches the observed data. The RMSE value indicates a reasonable level of predictive accuracy for the linear regression model, although there is room for improvement.

Discussion

Explanatory and Prediction Comparison

The main difference between the explanatory and prediction approaches lies in the goals of each problem. The explanatory problem focused on understanding relationships between electricity costs and household type, number of occupants, and number of bedrooms. In contrast, the prediction problem focused on careful variable selection to create a model that could predict electricity costs for households in Oregon.

To follow through on the goals of the explanatory problem there had to be careful attention to residual diagnostics and interpretations when determining the best fit for a model. I also had to evaluate p-values and coefficients to determine whether to retain interaction terms or not. In the prediction problem, cross validation played a big role when selecting the best subset selection for an accurate model.

Some of the challenges in the explanatory analysis included deciding which interaction terms to keep and comparing the different models. Removing terms that were not meaningful was necessary to improve the model's performance. In the prediction analysis, it was a challenge to find an optimal number of predictor variables. The k-10-fold cross validation was a key approach in selecting the right number.

Conclusion

In conclusion, it was discovered that people living in houses pay on average $19.47 more per month than people living in apartments. This was after adjusting for the number of bedrooms and household size.

The prediction analysis concluded with a 13-predictor variable model that would predict the cost of electricity in a household in Oregon. This model achieved a relatively low RMSE value of $66.93 per month which indicates a reasonable level of predictive accuracy for the linear regression model. I further expanded this analysis by including interaction terms to the base model and collected meaningful results. I obtained a lower RMSE of $66.89 with p-value < 0.001 providing a statistically significant improvement in model fit that included the interaction terms between NP and BDSP, and between HFL and RMSP.

APPENDIX

1.EXPLANATORY PROBLEM R CODE

```
#READ THE DATA
house <- read.csv("FINAL_OR_acs_house_occ.csv", stringsAsFactors=TRUE)
#VIEW AND SUMMARIZE DATA
df <- house
```

```
head(df)
summary(df)
#CREATE VARIABLE HA TO DISTINGUISH BETWEEN HOUSE, APT, AND OTHER
head(df$BLD, 20)
head(grepl("house", df$BLD), 20)
df$HA <- "other"
df$HA[which(grepl("house", df$BLD,
ignore.case = TRUE))] <- "house"
df$HA[which(grepl("apartment", df$BLD,
ignore.case = TRUE))] <- "apt"
dataHouse <- subset(df, HA!="other")
#SEE HA
dataHouse$HA <- factor(dataHouse$HA)
head(dataHouse$HA)
#SUMMARIZE HA
summary(dataHouse$HA)
#LOOK AT EXPLORATORY ANALYSIS
qplot(BDSP, ELEP, data = dataHouse, color = HA)
qplot(NP, ELEP, data = dataHouse, color = HA)
#fitHouseInt <- lm(ELEP ~ HA + BDSP + NP + HA:BDSP + HA:NP + BDSP:NP, data = #dataHouse)
#FIT MODEL WITH INTERACTIONS
fitHouseInt <- lm(ELEP ~ HA*BDSP + HA*NP + BDSP*NP, data = dataHouse)
summary(fitHouseInt)
#REMOVE NON SIGNIFICANT VARIABLE INTERACTION AND SUMMARIZE NEW MODEL
fitHouseInt2 <- lm(ELEP ~ HA*NP + BDSP*NP, data = dataHouse)
summary(fitHouseInt2)
#CHECK DIAGNOSTICS PLOT FOR FIT1
houseIntDiag <- augment(fitHouseInt, data = dataHouse)
qplot(.fitted, .resid, data = houseIntDiag)+ geom_hline(aes(yintercept=0))
qplot(HA,.resid, data = houseIntDiag)+ geom_hline(aes(yintercept=0))
qplot(BDSP,.resid, data = houseIntDiag)+ geom_hline(aes(yintercept=0))
qplot(NP,.resid, data = houseIntDiag)+ geom_hline(aes(yintercept=0))
#CHECK DIAGNOSTICS PLOT FOR FIT2
houseIntDiag2 <- broom::augment(fitHouseInt2, data = dataHouse)
qplot(.fitted, .resid, data = houseIntDiag2)+ geom_hline(aes(yintercept=0))
qplot(HA,.resid, data = houseIntDiag2)+ geom_hline(aes(yintercept=0))
qplot(BDSP,.resid, data = houseIntDiag2)+ geom_hline(aes(yintercept=0))
qplot(NP,.resid, data = houseIntDiag2)+ geom_hline(aes(yintercept=0))
#FIT MODEL WITHOUT INTERACTIONS
fitHouse <- lm(ELEP ~ HA + BDSP + NP, data = dataHouse)
summary(fitHouse)
#CHECK RESIDUAL PLOT FOR FIT WITH NO INTERACTION
houseDiag <- broom::augment(fitHouse, data = dataHouse)
qplot(.fitted, .resid, data = houseDiag)+ geom_hline(aes(yintercept=0))
qplot(HA,.resid, data = houseDiag)+ geom_hline(aes(yintercept=0))
qplot(BDSP,.resid, data = houseDiag)+ geom_hline(aes(yintercept=0))
qplot(NP,.resid, data = houseDiag)+ geom_hline(aes(yintercept=0))
#COMPARE MODELS USING ANOVA
anova(fitHouseInt, fitHouse)
#SUMMARIZE BEST MODEL COEFFICIENTS
summary(fitHouse)$coefficients
confint(fitHouse)
```

2. PREDICTION PROBLEM R CODE

```
#LOOK AT NA. what variables have NA?....ACR AND VALP
summary(dataHouse)
#PLOT NA to see relationships between variables and elep
qplot(ACR, ELEP, data = dataHouse)
```

```
qplot(ELEP, VALP, data = dataHouse)
### PREDICTION PROBLEM DATA ##
dataHouse2 <- subset(dataHouse, select = -c(ACR, SERIALNO, TYPE, BLD, VALP))
head(dataHouse2)
set.seed(001)
n <- nrow(dataHouse2)
valid <- sample(n,size = floor(0.20*n))
houseValid <- dataHouse2[valid,]
houseTrain <- dataHouse2[-valid,]
bigMod <- lm(ELEP ~ ., data = houseTrain)
length(bigMod$coefficients)
##BEST SUBSET##
regfitFull <- regsubsets(ELEP ~ ., data = houseTrain, nvmax = 37, really.big = TRUE)
regSummary <- summary(regfitFull)
regSummary
names(regSummary)
regSummary$rsq
#PLOT ADJUSTED R^2, CP, AND BIC
par(mfrow = c(2,2), mar = c(4,4,1,1))
plot(regSummary$adjr2, xlab = "# of Variables", ylab = "Adjusted RSq")
which.max(regSummary$adjr2)
points(24, regSummary$adjr2[24], col = "red", cex = 2, pch = 20)
plot(regSummary$cp,xlab = "# of Variables", ylab = "Cp")
which.min(regSummary$cp)
points(21, regSummary$cp[10], col = "red", cex = 2, pch = 20)
plot(regSummary$bic, xlab = "# of Variables", ylab = "BIC")
which.min(regSummary$bic)
points(13, regSummary$bic[13], col = "red", cex = 2, pch = 20)
#PLOT BIC
plot(regfitFull, scale="bic")
#MODEL WITH SMALLEST BIC
which.min(regSummary$bic)
coef(regfitFull, 13)
#CHOOSING AMONG MODELS WITH VALIDATION AND CROSS-VALIDATION
set.seed(10)
train <- sample(c(TRUE, FALSE), nrow(dataHouse2), replace = TRUE)
test <- (!train)
regfit.best <- regsubsets(ELEP ~ ., data = dataHouse2[train,], nvmax = 37)
#regfit.bestSummary <- summary(regfit.best)
#regfit.bestSummary
#MODEL MATRIX FROM TEST DATA
test.mat <- model.matrix(ELEP ~ ., data = dataHouse2[test,])
val.errors <- rep(NA, 37)
for(i in 1:37){
coefi <- coef(regfit.best, id=i)
pred <- test.mat[,names(coefi)] %*% coefi
val.errors[i] <- mean((dataHouse2$ELEP[test]-pred)^2)
}
val.errors
which.min(val.errors)
coef(regfit.best, 7)
#K-10 FOLDS
k <- 10
set.seed(100)
folds <- sample(1:k, nrow(dataHouse2), replace = TRUE)
cv.errors <- matrix(NA, k, 37, dimnames = list(NULL, paste(1:37)))
for(j in 1:k){
```

```
best.fit <- regsubsets(ELEP ~ ., data = dataHouse2[folds!=j,], nvmax = 37)
for(i in 1:37){
pred <- predict(best.fit, dataHouse2[folds==j,], id = i)
cv.errors[j,i] <- mean((dataHouse2$ELEP[folds==j]-pred)^2)
}
}
#MATRIX TO TEST MSE FOR ITH CROSS-VALIDATION FOLD FOR BEST J-VARIABLE MODEL.
mean.cv.errors <- apply(cv.errors, 2, mean)
mean.cv.errors
#PLOT MEAN CV ERRORS
par(mfrow = c(1,1))
plot(mean.cv.errors, type = "b", xlab = "# of Variables")
std.line <- min(mean.cv.errors) + (sd(mean.cv.errors)/sqrt(k))
abline(h=std.line)
reg.best <- regsubsets(ELEP ~ ., data = dataHouse2, nvmax = 37)
coef(reg.best, 12)
#RMSE
full.model <- lm(ELEP ~ NP + BDSP + FULP + GASP + HFL + RMSP + R60 + HA,
data = houseTrain)
houseValid$pred <- predict(full.model, newdata = houseValid)
(rmse <- with(houseValid, sqrt(mean((ELEP - pred)^2))))
summary(full.model)
# Add interaction terms
interaction.model <- lm(ELEP ~ NP * BDSP + FULP + GASP + HFL * RMSP + R60 + HA,
                        data = houseTrain)
# Predict and calculate RMSE
houseValid$pred_interaction <- predict(interaction.model, newdata = houseValid)
(rmse_interaction <- with(houseValid, sqrt(mean((ELEP - pred_interaction)^2))))
# Compare RMSE values using ANOVA.
# Look at the p-value: if it's small (e.g., < 0.05), the added interactions
# significantly improve the model.
anova(full.model, interaction.model)
```